JEE Mains 2020 Jan Chapter wise Question Bank

Indefinite Integration

01

Let $\int \frac{\cos x\, dx}{\sin^3 x(1+\sin^6 x)^{\frac{2}{3}}} = f(x)\ (1+\sin^6 x)^{\frac{1}{\lambda}} + c$ then find value of $\lambda f\left(\frac{\pi}{3}\right)$

माना $\int \frac{\cos x\, dx}{\sin^3 x(1+\sin^6 x)^{\frac{2}{3}}} = f(x)\ (1+\sin^6 x)^{\frac{1}{\lambda}} + c$, तो $\lambda f\left(\frac{\pi}{3}\right)$ का मान है–

(1) 4 (2) –2 (3)8 (4) –4

8th Jan Morning

Sol

(2)

$\sin x = t$

$\cos x\, dx = dx$

$$I = \int \frac{dt}{t^3(1+t^6)^{\frac{2}{3}}} = \int \frac{dt}{t^7\left(1+\frac{1}{t^6}\right)^{\frac{2}{3}}}$$

Put $1+\frac{1}{t^6} = r^3$ रखने पर $\Rightarrow \frac{dt}{t^7} = \frac{-1}{2}r^2 dr$

$$-\frac{1}{2}\int \frac{r^2\, dr}{r^2} = -\frac{1}{2}\ r + c = -\frac{1}{2}\left(\frac{\sin^6 x+1}{\sin^6 x}\right)^{\frac{1}{3}} + c = -\frac{1}{2\sin^2 x}(1+\sin^6 x)^{\frac{1}{3}} + c$$

$f(x) = -\frac{1}{2}\operatorname{cosec}^2 x$ and और $\lambda = 3$

$\lambda f\left(\frac{\pi}{3}\right) = -2$

02

Find integration $\int \frac{dx}{(x-3)^{6/7}.(x+4)^{8/7}}$

समाकलन $\int \frac{dx}{(x-3)^{6/7}.(x+4)^{8/7}} =$

(1) $\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}}+c$ (2) $7\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}}+c$ (3) $7\left(\frac{x-3}{x+4}\right)^{\frac{6}{7}}+c$ (4) $7\left(\frac{x+4}{x-3}\right)^{\frac{6}{7}}+c$

9th Jan Morning

Sol

(1)

$\int\left(\frac{x-3}{x+4}\right)^{\frac{-6}{7}}\frac{1}{(x+4)^2}dx$

Let $\frac{x-3}{x+4}=t^7$,

$\frac{7}{(x+4)^2}dx=7t^6dt$

$\int t^{-6}t^6dt=t+c$

03

$\int\frac{d\theta}{\cos^2\theta(\sec 2\theta+\tan 2\theta)} = \lambda\tan\theta + 2\log f(x) + c$ then ordered pair $(\lambda, f(x))$ is

$\int\frac{d\theta}{\cos^2\theta(\sec 2\theta+\tan 2\theta)} = \lambda\tan\theta + 2\log f(x) + c$ तब क्रमित युग्म $(\lambda, f(x))$ है–

(1) $(1, 1 + \tan\theta)$ (2) $(1, 1 - \tan\theta)$ (3) $(-1, 1 + \tan\theta)$ (4) $(-1, 1 - \tan\theta)$

9th Jan Evening

Sol

(3)

$$\int \frac{\sec^2\theta}{\frac{1+\tan^2\theta}{1-\tan^2\theta}+\frac{2\tan\theta}{1-\tan^2\theta}}\, d\theta$$

$$= \int \frac{\sec^2\theta(1-\tan^2\theta)}{(1+\tan\theta)^2} d\theta$$

$$= \int \frac{\sec^2\theta(1-\tan\theta)}{1+\tan\theta} d\theta$$

$\tan\theta = t \Rightarrow \sec^2\theta\, d\theta = dt$

$$= \int\left(\frac{1-t}{1+t}\right) dt = \int\left(-1+\frac{2}{1+t}\right)dt$$

$= -t + 2\log(1+t) + C$

$= -\tan\theta + 2\log(1+\tan\theta) + C$

$\Rightarrow \quad \lambda = -1$ and $f(x) = 1 + \tan\theta$